

Scalability & Future Plan

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Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Scalability & Future Plan

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	SQLite is intended for a lightweight prototype.	Limited concurrency and scaling options.	High
2	Direct external AI calls can add variable latency.	User wait time depends on API/network performance.	High
3	Prototype authentication is basic.	Production use would need stronger identity and authorization controls.	High
4	Generated content needs validation and safety controls.	AI output may require review before real-world use.	Medium

Scalability Plan

Aspect	Current State	Proposed Upgrade
User Load	Local prototype	Deploy with production server and horizontal scaling.
Data Storage	SQLite	Move to PostgreSQL or managed relational database.
Performance	Synchronous API calls	Use async/background processing, caching and monitoring.
Security	Environment secrets + basic admin session	Use secure identity provider, HTTPS, CSRF/session hardening, role-based access and secret management.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy hosted version and migrate database	Next release	Improved availability and multi-user support
Phase 3	Add progress tracking, plan history and richer analytics	Future iteration	Better continuity and user engagement
Phase 4	Mobile-friendly/PWA experience and advanced personalization	Longer term	Broader access and personalization